

CHE213 MASS TRANSFER
INNOVATION PROJECT

VARIABLE REFLUX CONTROL FOR CONSTANT DISTILLATE COMPOSITION

Group 21

Vipin Gora (241168)

Divy Kumar Pandey (240368)

Vishnuprasad (231167)

INTRODUCTION

- Batch distillation is a fundamental separation process where a fixed quantity of liquid mixture is charged into a reboiler, boiled, and the resulting vapors are continuously condensed and collected over time.
- However, unlike continuous column operations, where the feed concentration remains completely steady, in the batch distillation method, the concentration continuously depletes. As the more volatile component (ethanol) escapes into the vapor phase, the concentration left behind in the boiling kettle drops continuously.
- If a column is operated at a standard, fixed reflux ratio, the top product purity steadily degrades as the run progresses.
- Therefore, to maintain a high and constant product specification, the column's operating parameters cannot remain static; they must dynamically adapt to fight this inherent depletion.

OBJECTIVE

- Establish a baseline for product degradation at fixed reflux ($R=2$), then implement a dynamic feed-forward control strategy to maintain a constant ethanol product of 89%.
- To tackle the problem of reducing distillate purity by increasing the reflux ratio over time.
- To physically demonstrate that implementing a variable reflux operation maintains constant distillate purity over time.
- Develop a hybrid analytical method (Specific Gravity + Refractive Index) and simulate the required dynamic Reflux Ratio using a custom McCabe-Thiele numerical solver.

APPARATUS & SETUP

- Batch distillation column
- Reboiler (heating unit)
 - To heat the liquid mixture at the bottom
- Condenser with cooling water supply
 - To condense the ethanol-rich vapors
- Refractometer - For RI measurement
- Thermometers / temperature sensors
- Measuring cylinders and sample bottles
 - To collect and measure the samples
- Ethanol–water mixture
- Weighing machine - To measure the specific gravity
- Stopwatch - to measure time for 10 minute intervals



Batch distillation setup

INNOVATION ASPECTS

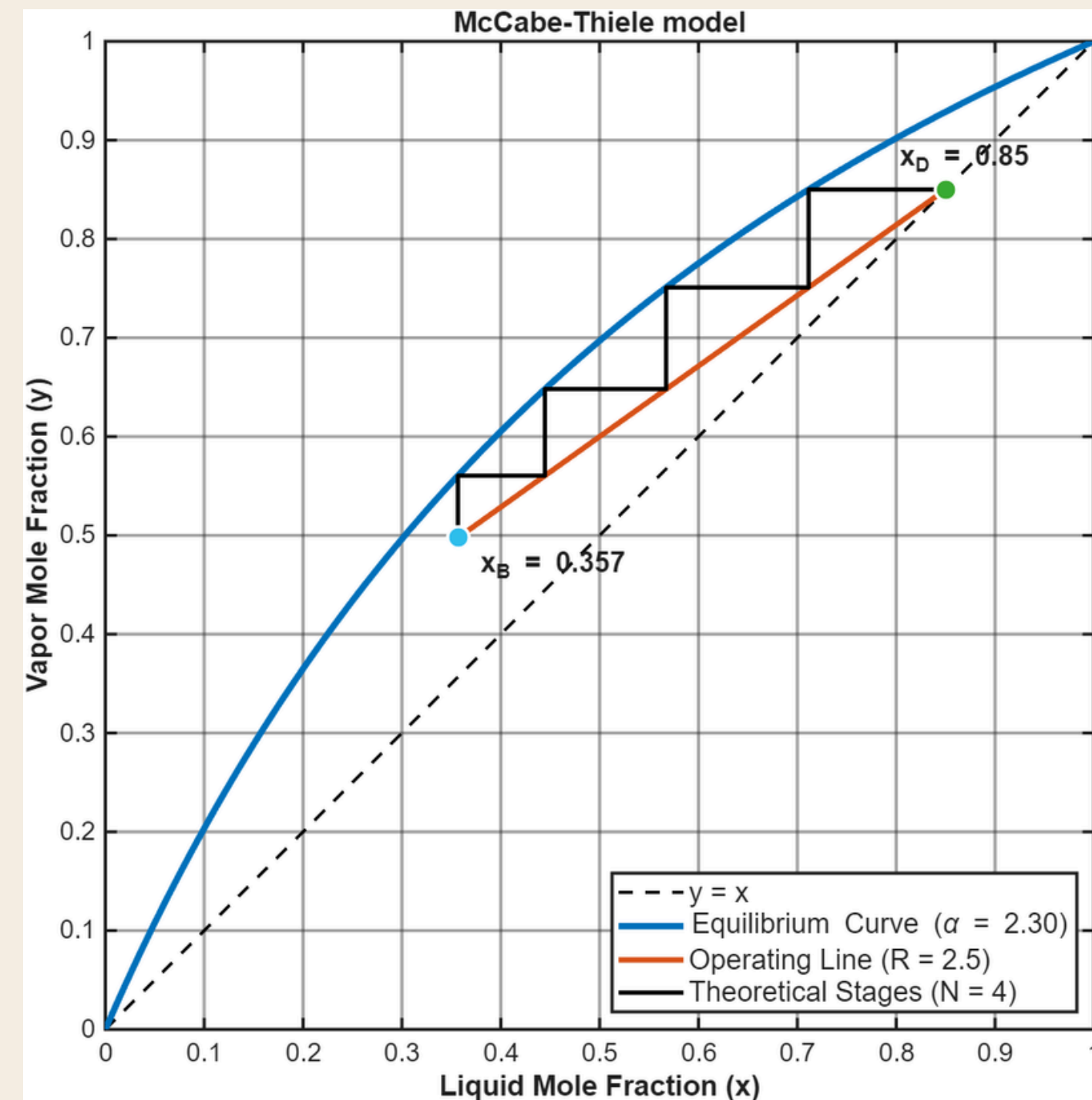
Overcome the degrading distillate concentration problem by increasing the reflux ratio over time.

Study the McCabe-Thiele plot and simulate the method to instantly find the required reflux ratio at a given time, using the bottom reboiler composition and the target distillate concentration

THEORETICAL BACKGROUND

McCabe-Thiele model:

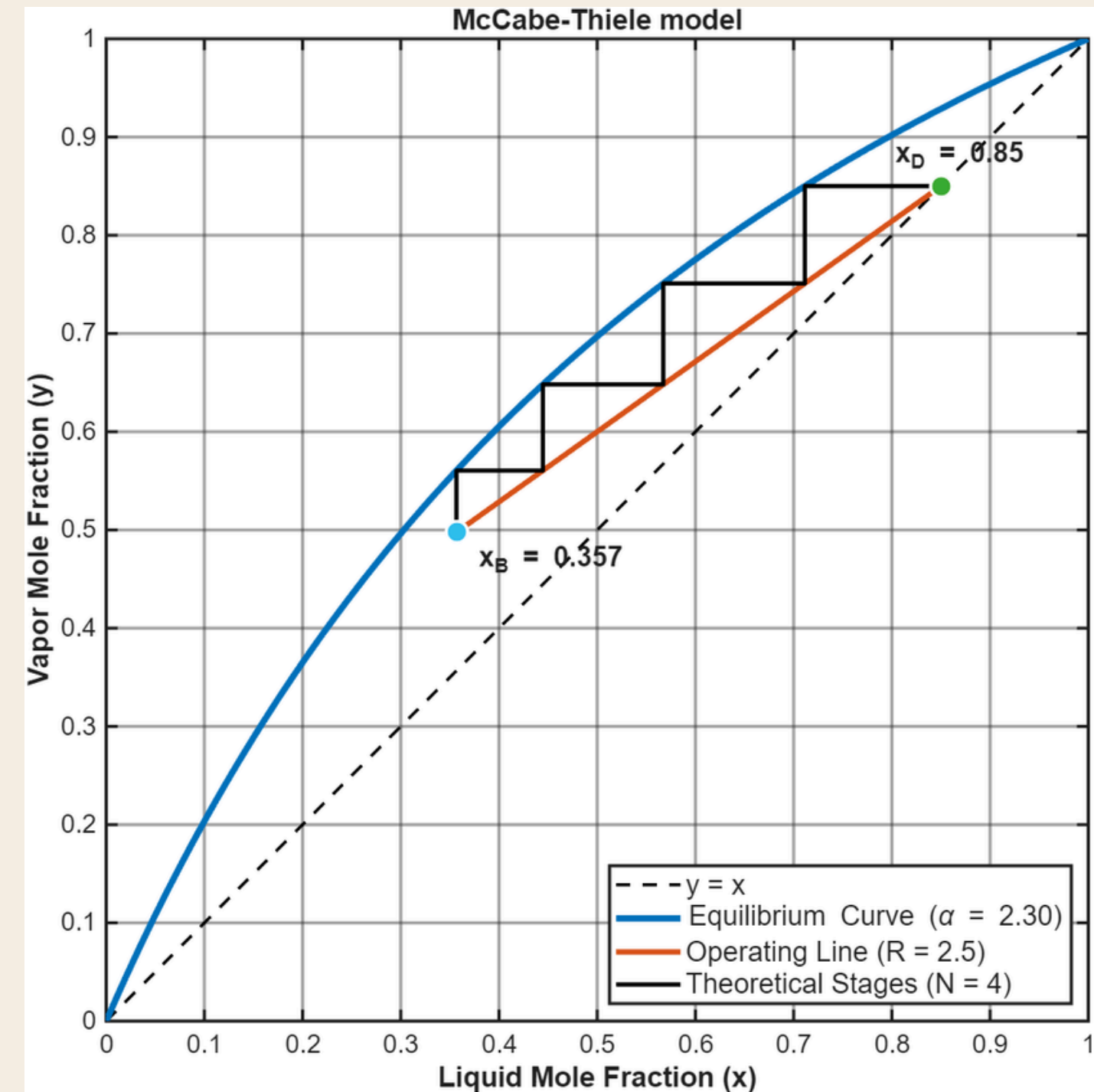
- Evaluates binary mixture separation using stages.
- System Setup: Modelled as 4 Theoretical Stages. (Calculated during experiment).
- The Operating Line Equation:
$$y = \frac{R}{R+1}x + \frac{x_p}{R+1}$$
- The slope of this line is strictly controlled by the Reflux Ratio (R)
- For a given x_B value, $y_B = \frac{\alpha x_B}{1 + (\alpha - 1)x_B}$ where, α is the relative volatility. ($\alpha = 2.3$), for ethanol and water.



McCabe-Thiele plot

THEORETICAL BACKGROUND

- A fixed Reflux Ratio (R) locks the operating line slope in place. As the reboiler concentration (x_B) depletes, for a constant number of plates, the distillate concentration (x_D) also drops.
- By actively increasing R , we increase the slope of the operating line.
- Pushing the operating line closer to the equilibrium curve grants the column more theoretical separation power, neutralising the reboiler concentration depletion.



METHODOLOGY

Setup and steady state:

- Charge the batch distillation column with a 0.4 mole percentage ethanol-water mixture in the reboiler.
- Start heating and allow the system to reach steady state under total reflux.
- Note the initial temperatures, RI and specific gravity values 30 minutes after the temperatures become stable.

1. Fixed partial reflux ratio run:

- Switch to partial reflux mode and fix the reflux ratio, say ($R=2$).
- Start collecting samples every 10 minutes from both top and bottom, and measure the RI and specific gravity values of those samples and note them.

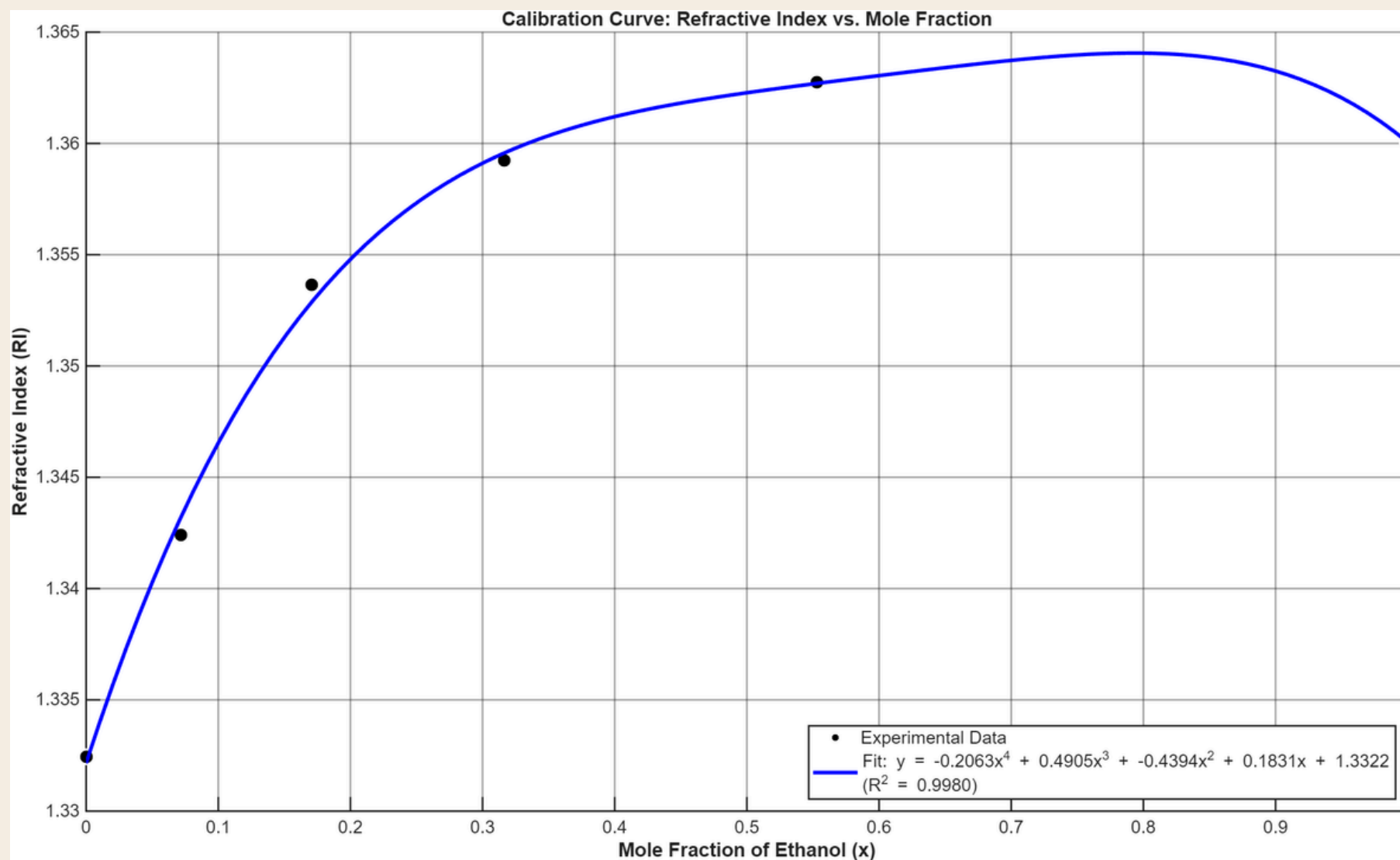
METHODOLOGY

2. Variable reflux ratio run

- Start by fixing the reflux ratio, say ($R=2.5$).
- Collect the samples after 10 minutes and measure the RI and specific gravity values.
- Immediately calculate the distillate concentration (x_D) and reboiler concentration (x_B) of ethanol.
- Put the values into MATLAB code to get the reflux ratio value required to achieve the target distillate concentration.
- Repeat the steps every 10 minutes.

CALIBRATION CURVES

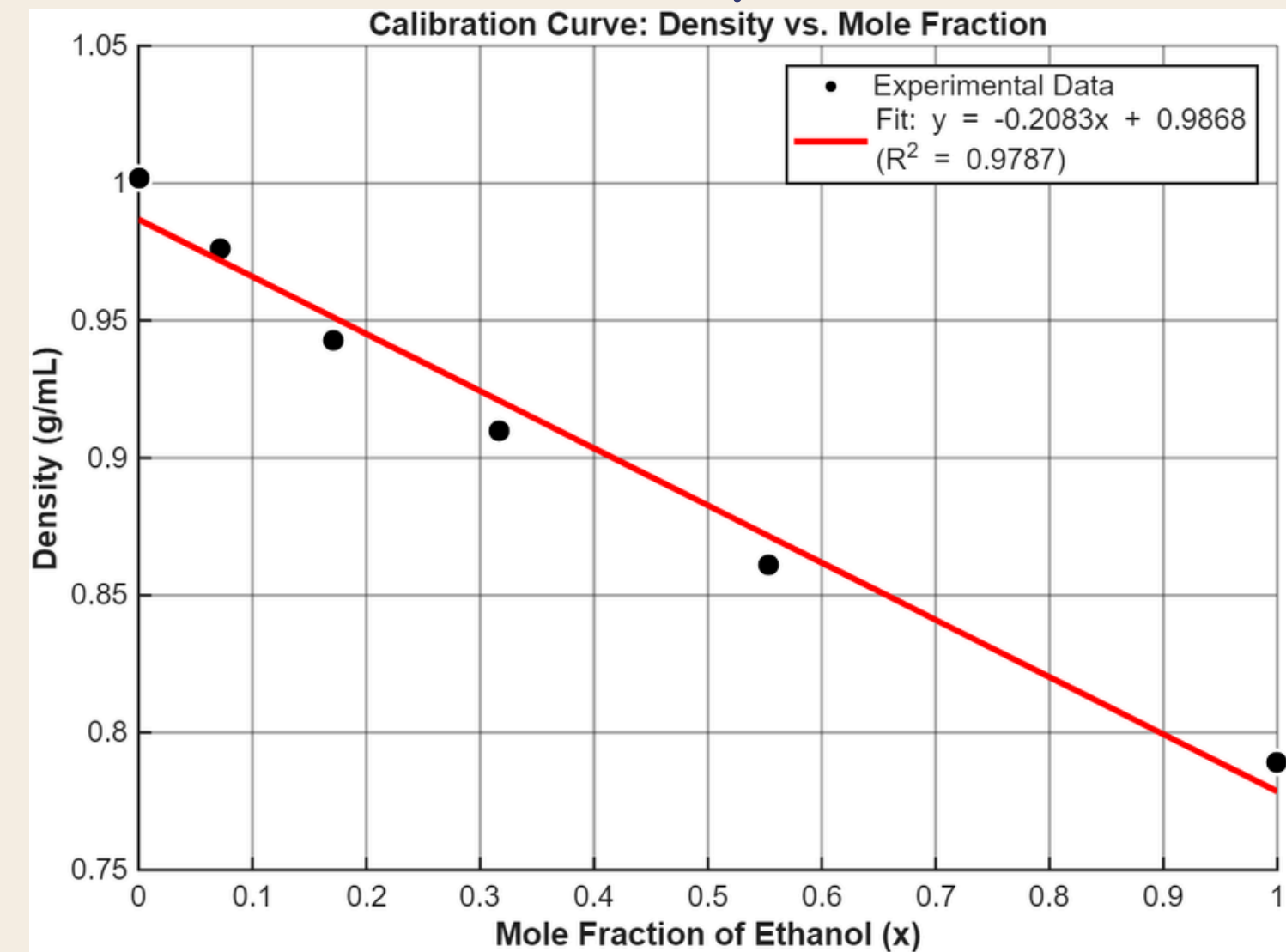
mole fraction vs RI calibration curve



Curve fit equation:

$$RI = -0.2063x^4 + 0.4905x^3 - 0.4394x^2 + 0.1831x + 1.3322$$

mole fraction vs density calibration curve



Curve fit equation:

$$y = -0.2083x + 0.9868$$

READINGS

Fixed reflux ratio (R = 2)

time (mins)	top temperature	bottom temperature	RI bottom	specific gravity top	xB	xD
10	77	85	1.36076	0.8131	0.369	0.834
20	77	85	1.36049	0.8154	0.355	0.823
30	77	85	1.36015	0.8188	0.339	0.807
40	77	85	1.35976	0.8228	0.322	0.787
50	77	85	1.35930	0.8262	0.305	0.771

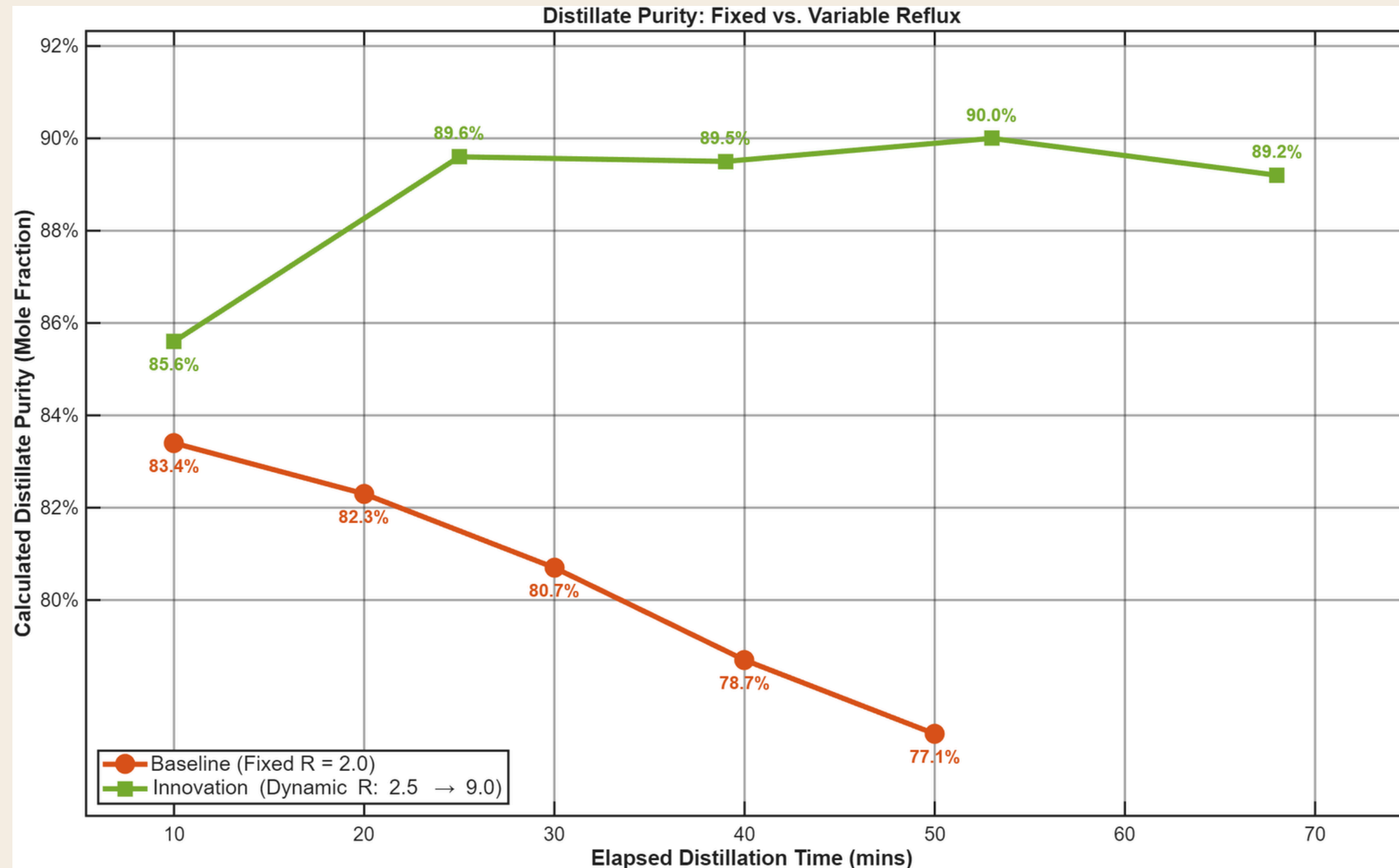
READINGS

Variable reflux ratio (samples colleted at the end of the time interveal)

time interval(mins)	top temperature	bottom temperature	RI bottom	specific gravity top	xB	xD	reflux ratio
0-10	77	85	1.36065	0.8085	0.362	0.856	2.5
15-25	77	85	1.36032	0.8002	0.342	0.896	5
29-39	77	85	1.36007	0.8003	0.332	0.895	6
43-53	77	85	1.35971	0.7994	0.320	0.900	8
58-68	77	85	1.35933	0.8011	0.307	0.894	9

COMPARISON GRAPH

Distillate concentrations for fixed and variable reflux ratios



CONCLUSIONS AND APPLICATIONS

- Successfully overcame the problem of degrading distillate composition.
- Proved that a simple time-stepped increase of the Reflux ratio can give a constant and required distillate concentration.
- Easily applicable method for industries, especially Pharmaceutical industries where a specific concentration of a chemical would be needed

THANK YOU

SPECIAL THANKS TO UOP STAFF